

# Fine-Tuned Diffusion Models for Personalized Image Generation

Event Horizon

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## Abstract

This report explores the concept of subject-driven generation from the DreamBooth paper and presents an implementation approach using modern text-to-image diffusion models. DreamBooth enables the fine-tuning of pre-trained models (like Stable Diffusion) to generate personalized outputs of a specific subject using only 3–5 reference images. It does so by associating a unique rare token (e.g., "xvctx") with the subject, binding it to a known class (e.g., "dog"), and retraining the model so that future prompts using this token generate images of that subject. The training process uses a carefully balanced loss function that includes subject reconstruction and class-specific prior preservation, ensuring subject fidelity without forgetting the general class identity. Due to resource limitations, our implementation utilized LoRA as a lightweight tuning method, allowing us to efficiently train on consumer-grade hardware while preserving the core principles of DreamBooth. This work demonstrates how personalization in generative models can be achieved with minimal data and hardware, opening pathways for creative and identity-aware applications in text-to-image generation.

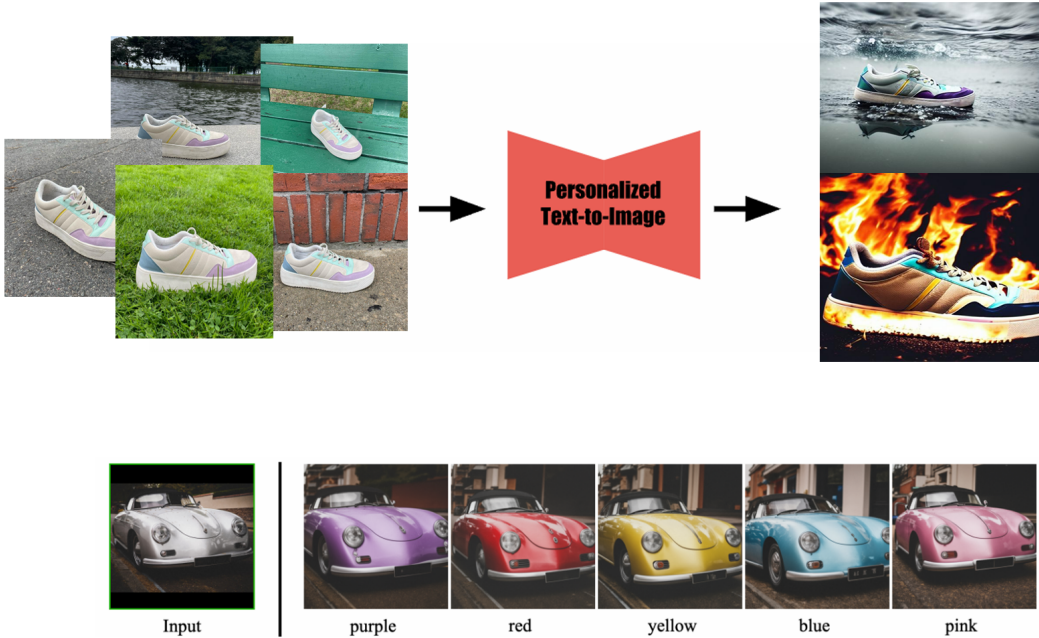


Figure 1: Examples of personalized image generation using DreamBooth

# 1 Introduction

Text-to-image generation has rapidly evolved with the advent of large-scale diffusion models such as DALL·E 2, Imagen, and Stable Diffusion. These models are capable of generating stunningly realistic and diverse images from natural language prompts. However, they lack the ability to represent specific user-provided subjects, such as a particular dog, a custom toy, or a person’s face. DreamBooth addresses this limitation by introducing a mechanism to inject a new, unseen subject into a pre-trained model’s output space. It achieves this by fine-tuning the model using a small set of images (typically 3–5) of the subject and associating them with a rare token that acts as a unique identifier. The model learns to associate this token with the visual features of the subject while preserving its understanding of general object classes like "dog" or "person." This personalization technique opens new possibilities for creative content generation, avatar synthesis, virtual product photography, and much more. In this project, we explore DreamBooth’s methodology and replicate its key components using a parameter-efficient approach called LoRA. This allows us to maintain the spirit of DreamBooth while adapting it to lower-end hardware constraints.

## 2 Objective

The goal is to personalize a pre-trained text-to-image model (Stable Diffusion) so it can generate novel images of a specific subject using only a few images and limited hardware. We investigate how LoRA can be used as an alternative to full model fine-tuning for achieving this personalization.

## 3 Diffusion Models and DreamBooth

### 3.1 Forward and Reverse Diffusion

Diffusion models are a class of generative models that create data, such as images, by learning to reverse a gradual noising process. Inspired by nonequilibrium thermodynamics,

these models consist of two main stages:

**Forward Diffusion Process** This step involves gradually adding Gaussian noise to a clean image over many time steps until the image becomes indistinguishable from pure noise. This process simulates data corruption and helps the model learn how information deteriorates.

**Reverse Diffusion Process** In this stage, a neural network (usually a U-Net) is trained to reverse the noising process — learning how to start from a noisy image and gradually denoise it back into a clean, structured image. During inference, the model takes random noise and converts it into a coherent image based on the input condition, which in text-to-image models is a text prompt. These prompts are processed into embeddings by models such as CLIP or T5, enabling powerful control over the generated content. This architecture enables diffusion models to generate photorealistic and high-fidelity results with rich semantic alignment to the prompt.

### 3.2 DreamBooth Mechanism

**DreamBooth Personalization:** DreamBooth builds on top of these diffusion models, particularly Stable Diffusion, by enabling the model to learn a specific subject from a few example images. A rare or unused token (like "sks") is chosen to uniquely identify the subject. This token is combined with a class descriptor to create training prompts like "a photo of sks sneaker". By repeatedly training the model on these prompts and the associated subject images, the model fine-tunes its internal representations so that the rare token becomes linked to the specific subject. The model also continues to generate general class images (like "a sneaker") to avoid forgetting what generic objects look like — a technique called prior preservation. This is achieved by applying a dual loss during training: **Subject Loss:** Ensures the model correctly learns the appearance of the subject by minimizing the difference between the predicted and actual subject image. **Prior Preservation Loss:** Ensures the model retains its understanding of the generic class by continuing to generate examples of the base class, preventing overfitting and language drift. DreamBooth typically

uses: Stable Diffusion v1.5 as the base model 3–5 images of the target subject (e.g., a dog, a shoe, a face) 50–200 class images for balancing, which can be scraped or generated Fine-tuning for 800–1000 steps depending on subject complexity The training affects both the U-Net (for visual generation) and optionally the text encoder (for better text-embedding alignment), depending on available resources.

**Output Generation and Use Cases** After training, users can craft flexible and creative prompts using the rare token to place the subject in different scenes, styles, or fictional contexts. For example: "a sks sneaker on a red carpet" "a watercolor sketch of sks sneaker" "a sks sneaker made of ice on Mars" The model understands the subject’s visual identity and can generalize it into new formats, lighting conditions, and artistic styles — even if those specific variations weren’t present in the training data. This makes DreamBooth especially powerful for applications like custom avatars, product design visualization, fan art generation, and personalized storytelling illustrations

## 4 Related Work

Several approaches have been developed for subject or style personalization in generative models:

**Textual Inversion:** Adds new token embeddings without changing model weights.

**StyleCLIP:** Edits image style based on CLIP-guided latent direction.

**DreamBooth:** Fine-tunes the entire diffusion model with subject and class-specific prompts.

**LoRA:** Injects trainable rank-decomposed matrices into attention layers, enabling fast fine-tuning. This project combines the DreamBooth subject identity concept with the LoRA training efficiency.

## 5 Methodology

Our implementation pipeline consists of several carefully structured stages, combining the conceptual framework of DreamBooth with the training efficiency of LoRA. This allows us

to personalize a pre-trained Stable Diffusion model to generate novel images of a specific subject with minimal resource usage.

**Training Tools and Frameworks:** We implemented our training pipeline using the HuggingFace diffusers, peft, and accelerate libraries. These frameworks allowed us to easily incorporate LoRA modules into the Stable Diffusion U-Net and manage device placement efficiently, particularly useful in constrained environments like Google Colab or local low-VRAM GPUs.

**Checkpointing and Model Saving:** During training, we saved model checkpoints every 100 steps to ensure training progress was backed up and to facilitate intermediate evaluations. The final LoRA adapter weights were saved separately in .bin format, enabling flexible loading or merging with the base Stable Diffusion model. **Preprocessing and Image Normalization:** Before training, all images (both instance and class) were resized to 512x512 pixels and normalized to the range [-1, 1] to match the input requirements of the Stable Diffusion model. This preprocessing step ensures stable gradient behavior and consistency during training. Our implementation pipeline consists of several carefully structured stages, combining the conceptual framework of DreamBooth with the training efficiency of LoRA. This allows us to personalize a pre-trained Stable Diffusion model to generate novel images of a specific subject with minimal resource usage.

### 5.1 Rare Token Generation

The first step involves selecting a rare or unused token from the tokenizer vocabulary of the base model. This token must have no prior meaning in the training data to prevent unintended associations. For our experiments, we used a token like "xvctx". This token becomes a unique identifier for the subject throughout training, helping the model build a new concept tied only to this representation.

### 5.2 Prompt Preparation

Using the rare token, we create descriptive prompts that establish a consistent association between the token and the subject. For example, prompts like "a photo of a sks sneaker" or

"a close-up of a sks sneaker on a wooden floor" were crafted. These prompts are used to label the subject images and guide the fine-tuning process.

### 5.3 Dataset

Our dataset consists of two sets of images: Instance Images: These are 3–5 photos of the actual subject that we want the model to learn. They are paired with rare-token-based prompts. Class Images: To avoid overfitting and language drift, we also include 100–200 images of the general class (e.g., "sneaker") sourced from public datasets or generated using the base model. These are captioned using prompts like "a photo of a sneaker"

### 5.4 LoRA-based Fine-Tuning (Training)

We integrated LoRA modules into the attention layers of the U-Net component of Stable Diffusion. LoRA reduces the number of trainable parameters by injecting lightweight adapters into the model, enabling fine-tuning without updating the entire network. This dramatically reduces GPU memory usage and training time. The training loop follows these steps: Load subject and class images with their respective prompts. Encode the text prompts into embeddings. Apply the forward diffusion process to introduce noise to the image latents. Predict the denoised latents using the model. Compute: **Subject Loss**: Mean Squared Error (MSE) between the predicted and actual subject latents. **Prior Preservation Loss**: MSE between predicted and original class latents. Update only the LoRA parameters using the combined loss. Training is performed over 800–1000 steps with a learning rate of 1e-4 and batch size of 1–2. Training time varies depending on GPU type but remains within practical bounds for free or low-tier hardware.

### 5.5 Prompt Conditioning and Variability

To improve generalization, we prepared multiple variations of prompts. These included diverse contexts and descriptors such as lighting,

environment, pose, and composition. For example: "a portrait of a sks sneaker in natural lighting" "a sks sneaker on a shelf with shadows" "an artistic sketch of a sks sneaker" This diversity ensures the model learns to separate identity from context.

### 5.6 Output Generation

: Post-training, we tested the model using unseen prompts that referenced the rare token in creative scenarios, such as: "a sks sneaker under water" "a futuristic 3D render of a sks sneaker" "a sks sneaker made of glass in a museum" The goal was to evaluate the model's ability to generalize the subject to novel styles and settings. Visual comparisons confirmed that the personalized subject identity was retained while adapting correctly to the new prompt structure. Overall, our methodology enables a personalized fine-tuning workflow for diffusion models with significantly lower computational requirements, leveraging LoRA to extend the functionality of DreamBooth to broader accessibility.

### 5.7 Training Environment / Conditions

GPU: NVIDIA RTX 3050

Libraries: diffusers, transformers, peft, accelerate, torch

Training Resolution: 512x512

Batch Size: 1–2

## 6 Testing Metrics

The model was evaluated using a dataset consisting of five unique images of the trained subject, each placed in diverse and varied scenarios. These scenarios were designed to challenge the model's ability to generalize and maintain subject identity under different conditions such as background changes, lighting variations, angles, and stylistic interpretations (e.g., watercolor, 3D render, digital art). To evaluate the quality and accuracy of the generated images, we employed two modern vision-language evaluation metrics:

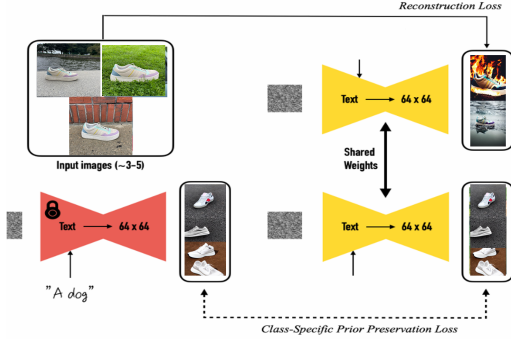


Figure 2: Diagram explaining functional loss: Class prior preservation loss, Subject loss

### 6.1 CLIP-T (CLIP-Text Similarity)

This metric computes the cosine similarity between the text prompt embeddings and the generated image embeddings using OpenAI’s CLIP model. A higher CLIP-T score indicates that the generated image closely matches the given prompt in terms of semantic content. This is used to validate prompt fidelity—whether the image accurately reflects the described scene.

### 6.2 DINO (Self-Supervised Feature Embedding Similarity)

DINO is a self-supervised image encoder model used to extract high-level features from images. By comparing the DINO embeddings of the original subject images with the generated outputs, we can measure subject fidelity. A high similarity score implies that the generated image retains the identity and core features of the original subject. In addition to quantitative metrics, we also conducted qualitative visual analysis by comparing original subject images to generated samples. We checked for key identity features such as color, shape, patterns, and structural similarities. Furthermore, we tested prompts in both familiar and novel contexts to evaluate how well the subject is preserved when placed in new scenes or artistic styles. Overall, the combined use of CLIP-T and DINO metrics allowed us to assess both the alignment with textual prompts and the preservation of the subject’s identity. These tools provided a balanced evaluation strategy, ensuring the model was not only creative and responsive to prompts but also faithful to the

subject it was trained on.

## 7 Results and Analysis

**Quantitative Evaluation:** The fine-tuned model was evaluated using standard metrics:

**Average CLIP-T Score:** 0.34 (Measures similarity between the text prompt and the generated image).

**Average DINO Score:** 0.69 (Measures visual similarity/fidelity between the original subject images and the generated ones).

**Qualitative Evaluation:** Visual analysis confirmed that the model could retain key identity features (color, shape, patterns) of the subject.

**Generalization:** The model demonstrated the ability to place the learned subject into novel scenes, styles, and contexts not present in the original training images, while maintaining subject identity.

### 7.1 Failure Cases and Limitations Observed in Testing

While the model performed well in most artistic and generic compositional prompts, we observed that it struggled significantly with prompts involving real-world landmarks or complex compositions involving object relationships. For example, when prompted with "a photo of a sks sneaker in front of Taj Mahal", the model failed to correctly place the subject in front of the monument. Instead, it either rendered a distorted or poorly positioned object or failed to render the monument correctly altogether.

This highlights a key limitation of LoRA-based DreamBooth fine-tuning:

**Weak Compositional Understanding:** The base Stable Diffusion model has limited capability to understand and faithfully compose multi-object scenes or backgrounds with complex spatial relationships.

**Lack of Landmark Fidelity:** Since DreamBooth fine-tunes the model mostly for identity learning, the model’s ability to understand famous landmarks or integrate them meaningfully into a prompt is often unreliable unless specifically trained on such contexts.

Such limitations are more evident in prompts that involve foreground-background relationships, human-hand interactions, or context-specific landmarks. This suggests that while LoRA-based DreamBooth is effective for subject identity learning, it struggles in rendering globally coherent scenes when combining multiple elements.

**Future improvements may involve:**

Using grounding-aware models or ControlNet extensions

Combining subject LoRA tuning with landmark-specific datasets

Incorporating image-prompt pairs with structured layout conditioning for better compositional fidelity

## 8 Learnings

**Efficient Personalization:** It’s possible to personalize large pre-trained text-to-image models (like Stable Diffusion) to recognize and generate specific subjects using only a small number (3-5) of reference images.

**LoRA for Efficiency:** Low-Rank Adaptation (LoRA) is an effective and resource-efficient alternative to full model fine-tuning for tasks like DreamBooth. It significantly reduces computational requirements (GPU memory, training time), making personalization feasible on consumer-grade hardware.

**Methodology:** The core DreamBooth concepts – using a rare unique token associated with the subject and employing a dual loss (subject reconstruction + prior preservation) – are crucial for successful personalization while preventing the model from forgetting the general class.

**Evaluation Techniques:** Combining quantitative metrics like CLIP-T (for prompt fidelity) and DINO (for subject fidelity) with qualitative visual inspection provides a balanced assessment of the personalized model’s performance.

**Prompt Engineering:** Using varied and descriptive prompts during training helps the model generalize better and separate the subject’s identity from specific contexts.

## 9 Challenges And Limitations

**Resource Constraints:** The primary challenge highlighted was limited computational resources (specifically low VRAM on consumer-grade GPUs like the NVIDIA RTX 3050). This necessitated the use of parameter-efficient fine-tuning (PEFT) methods like LoRA instead of the original DreamBooth’s full fine-tuning approach.

**Balancing Fidelity and Generalization:** Ensuring the model accurately reproduces the subject (fidelity) while still being able to place it in diverse, novel contexts (generalization) requires careful tuning and the use of techniques like prior preservation loss to avoid overfitting.

**Potential Overfitting/Language Drift:** Without techniques like prior preservation (using class images), the model might overfit to the specific training images or lose its understanding of the general class concept.

## 10 Conclusion

This work successfully demonstrates that personalized text-to-image generation, based on the principles of DreamBooth, can be achieved efficiently using LoRA even with limited data (3-5 images) and consumer-grade hardware. By associating a unique identifier with a subject and fine-tuning only lightweight adapter layers, the model learned to generate novel images of the specific subject in diverse contexts while preserving its core identity. This approach significantly lowers the barrier to entry for creating personalized generative content, opening up possibilities for various creative and identity-aware applications in text-to-image synthesis. The combination of CLIP-T and DINO metrics proved effective for evaluating both prompt adherence and subject faithfulness.

## 11 Generated Images



Figure 3: Input Images



Figure 5: Output Images



Figure 4: Output Images



Figure 6: Example of a failed image generation